

PDF Dependency Check

```
. * # Data Overview
. summarize price mpg weight length, separator(0)
Variable | Obs Mean Std. dev. Min Max
-----+-----
price | 74 6165.257 2949.496 3291 15906
mpg | 74 21.2973 5.785503 12 41
weight | 74 3019.459 777.1936 1760 4840
length | 74 187.9324 22.26634 142 233

. tabstat price mpg weight, by(foreign) statistics(n mean sd median min max) columns(statistics)
Summary for variables: price mpg weight
Group variable: foreign (Car origin)
```

foreign	N	Mean	SD	p50	Min	Max
Domestic	52	6072.423	3097.104	4782.5	3291	15906
	52	19.82692	4.743297	19	12	34
	52	3317.115	695.3637	3360	1800	4840
Foreign	22	6384.682	2621.915	5759	3748	12990
	22	24.77273	6.611187	24.5	14	41
	22	2315.909	433.0035	2180	1760	3420
Total	74	6165.257	2949.496	5006.5	3291	15906
	74	21.2973	5.785503	20	12	41
	74	3019.459	777.1936	3190	1760	4840

```
. * # Regression Analysis
. regress price mpg weight length i.foreign
Source | SS df MS Number of obs = 74
-----+----- F(4, 69) = 21.01
Model | 348708940 4 87177235 Prob > F = 0.0000
Residual | 286356456 69 4150093.57 R-squared = 0.5491
-----+----- Adj R-squared = 0.5230
Total | 635065396 73 8699525.97 Root MSE = 2037.2
```

price	Coefficient	Std. err.	t	P> t	[95% con f. interval]	
mpg	-13.40719	72.10761	-0.19	0.853	-157.2579	130.4436
weight	5.716181	1.016095	5.63	0.000	3.689127	7.743235
length	-92.48018	33.5912	-2.75	0.008	-159.4928	-25.46758
foreign						
Foreign	3550.194	655.4564	5.42	0.000	2242.594	4857.793
_cons	5515.58	5241.941	1.05	0.296	-4941.807	15972.97

```
. estimates store base
. * ## Margins
. margins foreign, atmeans
```

```
Linear prediction, predict()
At: mpg          = 21.2973 (mean)
weight          = 3019.459 (mean)
length         = 187.9324 (mean)
0.foreign      = .7027027 (mean)
1.foreign      = .2972973 (mean)
```

	+	Delta-method					
		Margin	std. err.	t	P> t	[95% con f. interval]	
foreign							
Domestic		5109.794	306.6837	16.66	0.000	4497.977	5721.611
Foreign		8659.987	517.9058	16.72	0.000	7626.794	9693.181

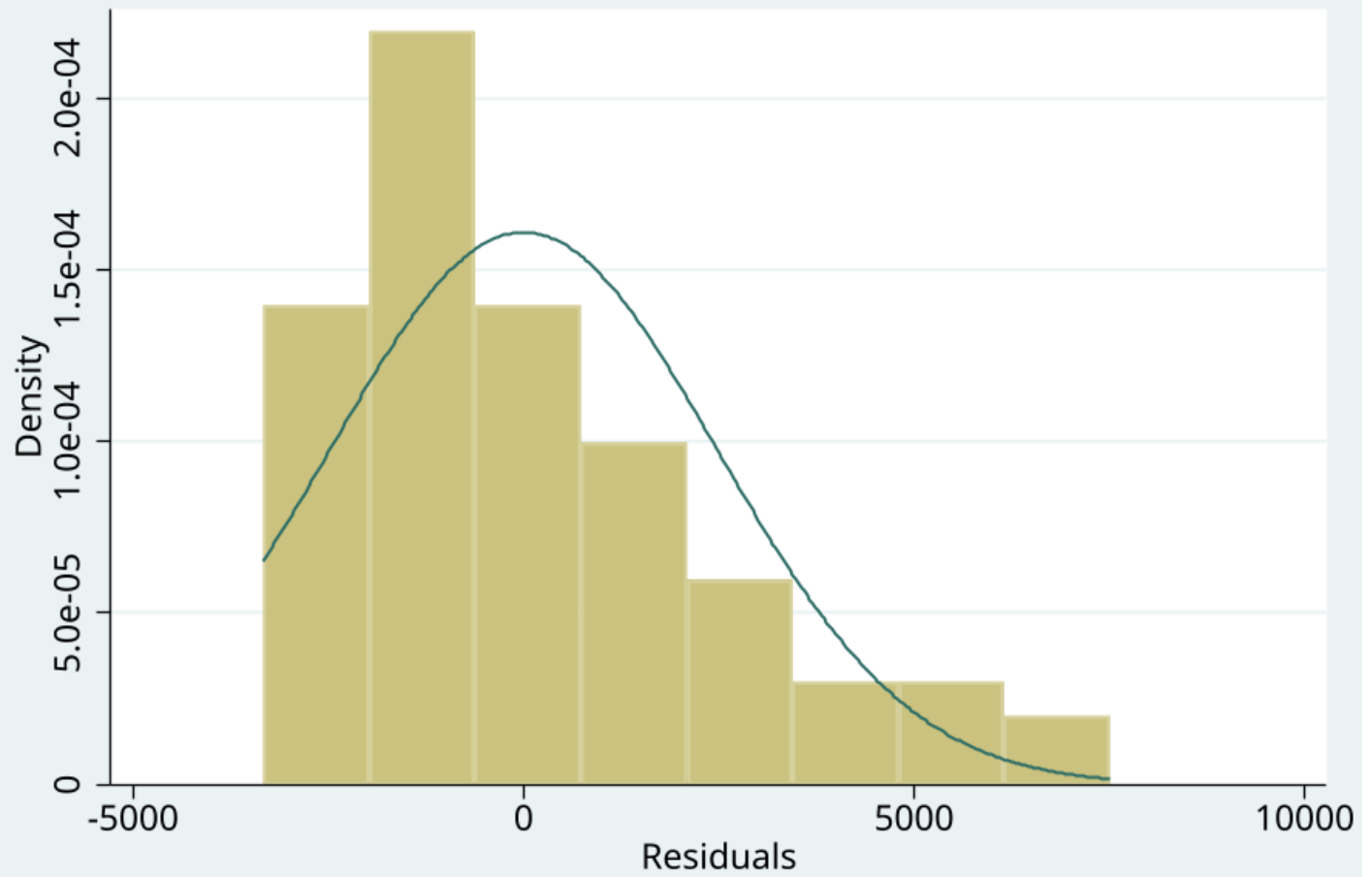
Key	
<i>frequency</i>	
<i>row percentage</i>	

		Repair record 1978				
Car origin	1	2	3	4	5	Total
Domestic	2	8	27	9	2	48
	4.17	16.67	56.25	18.75	4.17	100.00
Foreign	0	0	3	9	9	21
	0.00	0.00	14.29	42.86	42.86	100.00
Total	2	8	30	18	11	69
	2.90	11.59	43.48	26.09	15.94	100.00

```
. * # Diagnostics
. quietly regress price mpg weight
. predict double resid, residuals
. histogram resid, normal ///
>     title("Residual Distribution") ///
>     xtitle("Residuals") ytitle("Density")
(bin=8, start=-3332.4617, width=1354.9264)

. graph export "logdoc/demo/residuals.png", replace width(900)
```

Residual Distribution



file logdoc/demo/residuals.png written in PNG format

```
. capture graph close _all  
. drop resid
```